ASSIGNMENT - II REPORT

Air Quality Monitoring Using Official & Community Sensors

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List of abbreviations

SDG: Sustainable Development Goals

VHR: Very High Resolution

UN: United Nations

GIS: Geographic Information System

GSO: Generic Slum Ontology

DTM: Digital Terrain Model

GDAL: Geospatial Data Abstraction Library

OSM: Open Street Maps

NDVI: Normalised Difference Vegetation Index

1. Introduction

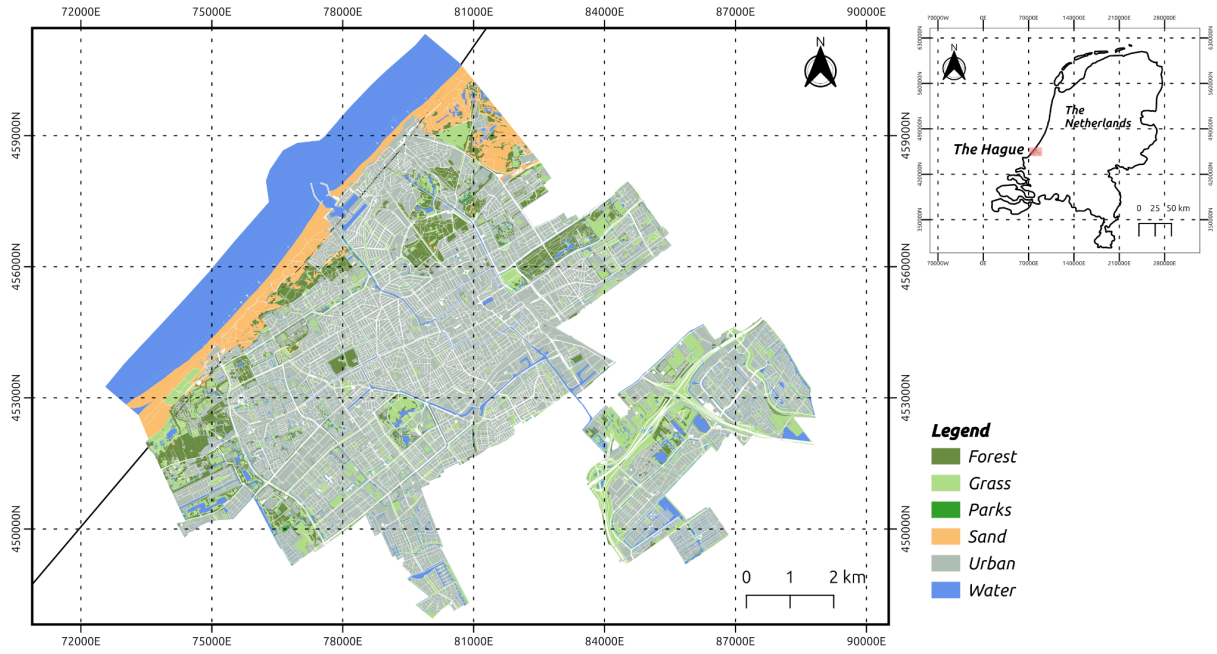
Air pollution poses significant risks to human health and the environment. It contributes to numerous respiratory and cardiovascular diseases, reduces quality of life, and leads to preventable deaths. Fine particulate matter (PM2.5) is the primary pollutant of concern, representing the greatest health risk. These particles are often formed in the atmosphere through complex reactions of gaseous oxides, such as sulfur and nitrogen oxides, which are primarily emitted from domestic heating, industrial activities, and road transport.

Although air quality in Europe has improved over the past decade, air pollution remains the leading environmental health hazard in the region. Cities in the Netherlands are no exception. According to the European Environment Agency (EEA) 2023 report, the primary contributors to particulate matter pollution in the Netherlands are the residential, commercial, and institutional sectors, followed by manufacturing, extractive industries, and road transport. Research estimates an average reduction in life expectancy of 6 to 12 months due to air pollution in the Netherlands (EEA, 2007).

In 2023, cities such as Waalwijk, The Hague, Oirschot, and Mijnsheerenland were among the most polluted in the Netherlands, with PM2.5 levels exceeding 2–3 times the acceptable thresholds for much of the year. Conversely, cities like Groningen, Munnikeijl, and Wieringerwerf enjoyed relatively clean air throughout the year. Air quality in the Netherlands is monitored continuously at approximately 100 locations by organizations such as the National Institute for Public Health and the Environment (RIVM), the GGD Amsterdam, and DCMR Rotterdam. In addition to these 24/7 monitoring stations, non-continuous mobile sensors, known as community sensors, are also deployed through platforms like the Samen Meten Data Portal.

This report evaluates the air quality of one Dutch city using data from both official monitoring stations and community-based sensors. It assesses the reliability of these measurements and proposes an optimized layout plan for sensor placement to maximize coverage, with a particular focus on monitoring air quality around schools.

1.2 Study Area



Study Area: The Hague, The Netherlands

Centroid Coordinates: 79982E, 454320N
Data Source: OpenStreetMap (OSM), Top10NL (pdok.nl) data retrieved on 12th Jan, 2025

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

This case study focuses on The Hague, a municipality and the capital of South Holland province in the Netherlands. It is the third-largest city in the country, with a population exceeding half a million. While Amsterdam is the official capital of the Netherlands, The Hague serves as the administrative center and the seat of government.

The Hague is also one of the most densely populated municipalities in the Netherlands, with a population density of approximately 6,868 people per square kilometer, spread over an area of 9,820 hectares. The majority of this area is built-up, leaving limited green cover. The western part of the city is characterized by coastal dunes and some green spaces.

Traffic congestion is a notable issue in The Hague. According to the 2023 TomTom Traffic Report, residents spent an average of 1.5 hours more traveling within the city compared to the previous year. This dense urban setup and high traffic volume significantly impact the environment, contributing to elevated pollution levels.

Air pollution in The Hague is a pressing concern, with concentrations of pollutants frequently exceeding both European standards and World Health Organization (WHO) guidelines. This highlights the critical need for active monitoring and targeted interventions to identify and mitigate air quality hotspots.

1.3 Objectives

1. To evaluate the current reliability of the community sensor network in The Hague.

2. To develop an optimized placement strategy for air quality sensors to ensure maximum coverage of city schools.
3. To assess the improvement in reliability following the installation of new sensors.

2. Material & Methods

2.1 Data Acquisition

To quantify air quality, this study focuses on two pollutants: PM2.5 (particulate matter <2.5 µm) and PM10 (particulate matter <10 µm). Particulate matter consists of aerosol particles emitted primarily through anthropogenic activities or formed in the atmosphere via chemical reactions involving precursor gases like NOx and SOx. These pollutants were chosen for their severe and hazardous impact on human health. PM2.5 is particularly dangerous as it penetrates deeply into the lungs. For perspective, an average human hair is approximately 70 µm in diameter—30 times larger than a PM2.5 particle and 10 times larger than PM10.

The World Health Organization (WHO) has established guidelines for daily and annual limits for PM2.5 and PM10 emissions. The European Union (EU) also sets its own standards, which are more flexible but broadly align with WHO recommendations.

Pollutant	**WHO Limits (µg/m³)**	**EU Limits (µg/m³)**
PM2.5 (Annual)	5	25
PM2.5 (24-Hour)	15	Not Specified
PM10 (Annual)	15	40
PM10 (24-Hour)	45	50 (max 35 exceedances/year)

Table 1 Description of WHO and EU limits on PM10 and PM2.5 concentrations daily and annually.

2.1.1 Official Air Quality Monitoring

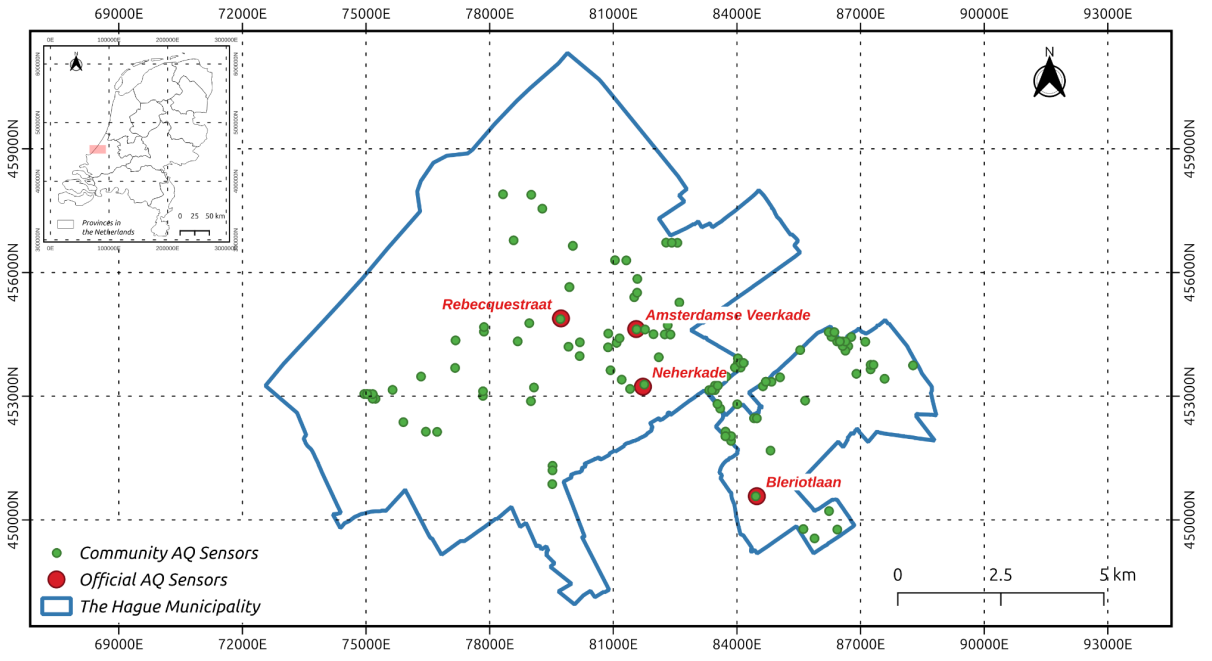
In the Netherlands, air quality is monitored through a collaboration of national agencies, including the National Institute of Public Health and the Environment (RIVM), GGD Amsterdam, the Environmental Protection Agency (EPA), and DCMR Rotterdam. These organizations utilize a combination of high-precision reference-grade instruments and low-cost sensors. Data is accessible via the air quality portal www.luchtmeetnet.nl in real-time and as archival datasets.

As of January 2025, there are approximately 101 official monitoring stations across the Netherlands, of which four are located in The Hague. All four measure PM10, while only two measure PM2.5. Sensor metadata and coordinates were retrieved using the **luchtmeet API**, while measurement data for the entire year was sourced from a different archival data portal due to the high API request limit and slow response times. A total of six years of hourly data (2019–2024) was collected to ensure sufficient temporal coverage, even accounting for gaps.

Sensors	Sensor ID	Pollutants	Type	Organisation
Rebecquestraat	NL10404	PM10, PM2.5	Municipal	RVIM
Amsterdamse Veerkade	NL10445	PM10	Traffic	RVIM
Bleriot Laan	NL10446	PM10	Municipal	RVIM
Neherkade	NL10450	PM10, PM2.5	Traffic	RVIM

2.1.2 Community Air Quality Monitoring

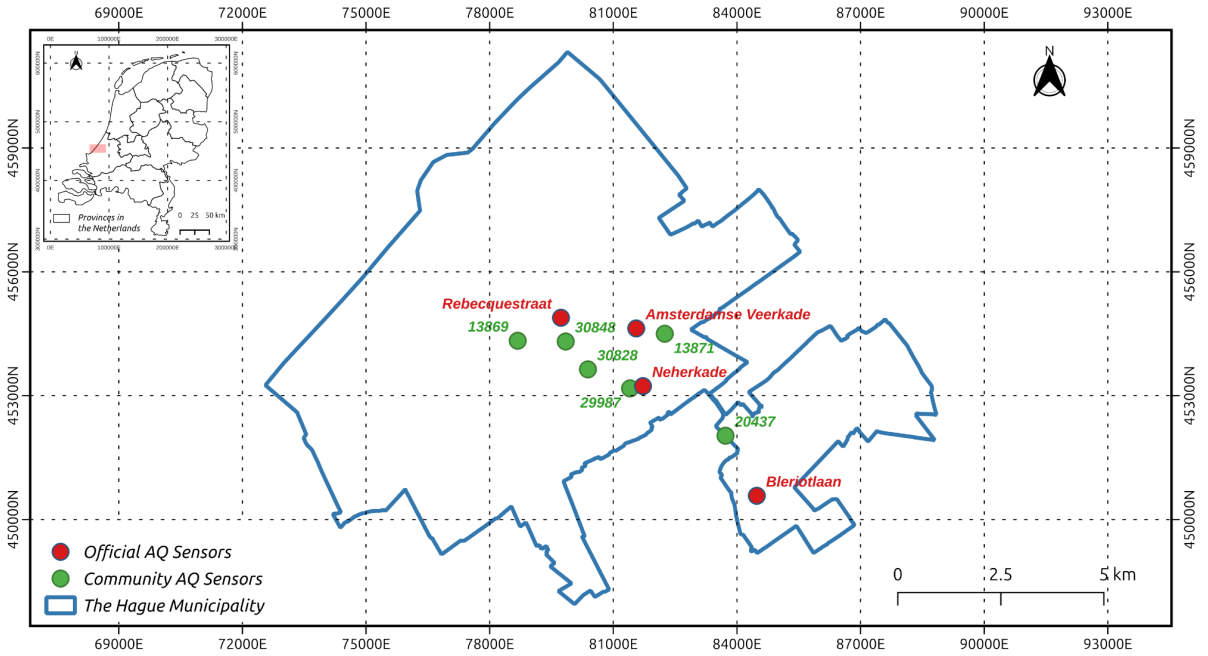
The Netherlands also has an extensive network of low-cost sensors maintained by the community, supported and calibrated by RIVM under the “Samen Meten” initiative. As of January 2025, approximately 19,584 community sensors operate across the country, with 140 located within the study area. By applying a proximity threshold of 500 m (and later extending to 1.5 km to address data gaps), 10 sensors were identified and stored in a shapefile with CRS EPSG:28992. For each sensor, six years of measurement data (2019–2024) was downloaded and saved as CSV files for further analysis.



Official & Community Air Quality Sensors

Centroid Coordinates: 79982E, 454320N
Data Source: Lucht MeetNet (luchtmeetnet.nl), Samen Meten (samenmeten.nl), OpenStreetMap (OSM)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841



Selected Official & Community Air Quality Sensors

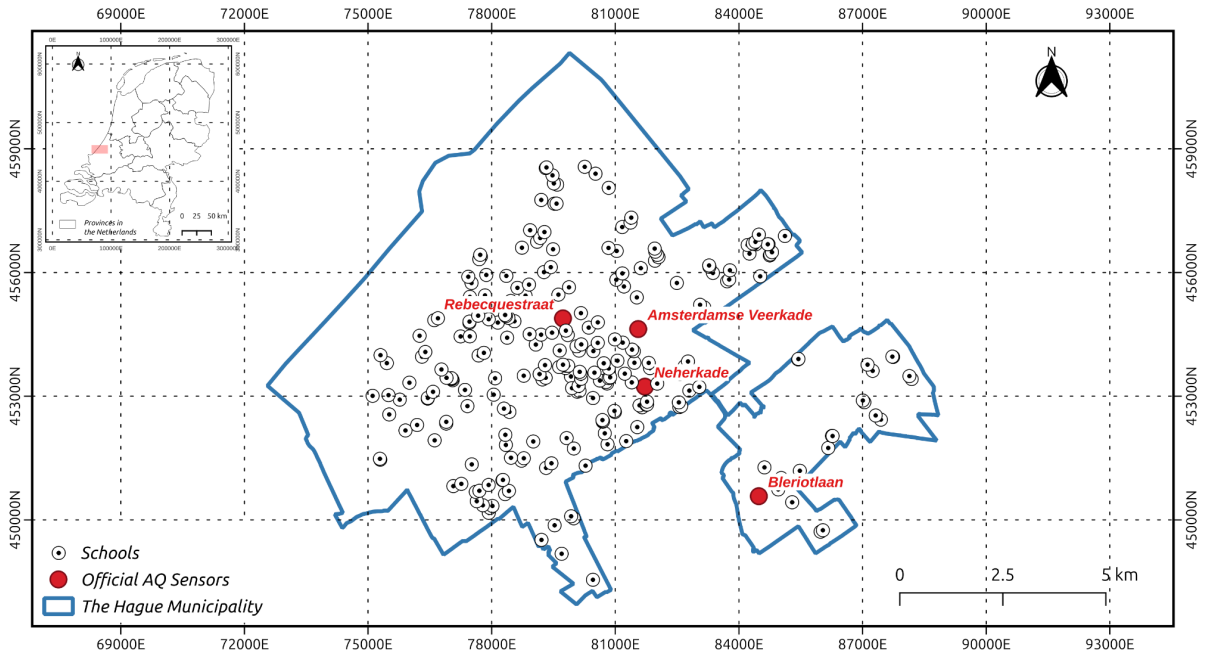
Centroid Coordinates: 79982E, 454320N
Data Source: Lucht MeetNet (luchtmeetnet.nl), Samen Meten (samenmeten.nl), OpenStreetMap (OSM)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

2.1.3 Schools

School locations within the study area were extracted from OpenStreetMap (OSM) using the tag “amenity=school.” A total of 251 schools were identified and stored as a shapefile with

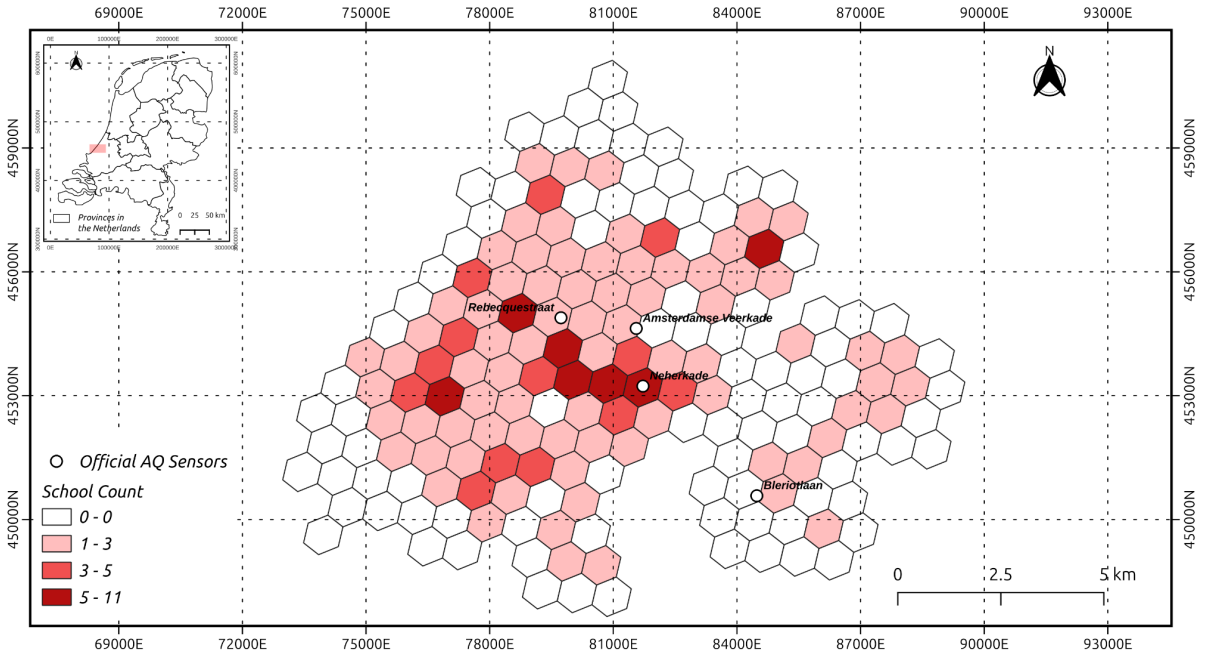
CRS EPSG:28992



Official Air Quality Sensors & Schools

Centroid Coordinates: 79982E, 454320N
Data Source: Lucht MeetNet (luchtmeetnet.nl), OpenStreetMap (OSM)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841



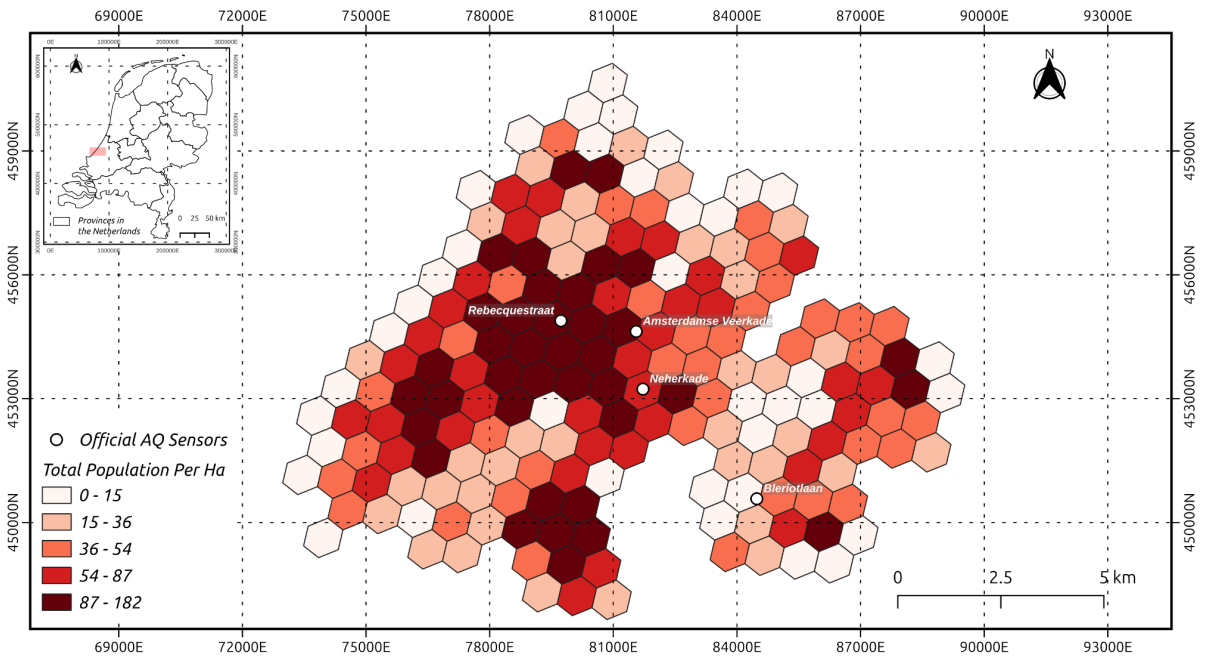
Distribution of Schools in 500m Hexagonal Grid Cells

Centroid Coordinates: 79982E, 454320N
Hex Grid: ~500m (Side length)
Data Source: Kontur Population Dataset (kontur.io/portfolio/population-dataset)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

2.1.4 Population

Population data was acquired from Kontur Population. It is represented by H3 hexagons with population counts at 400m resolution. The dataset is primarily developed using Global Human Settlement Layer (GHSL) but has also integrated Facebook population data (HRSL), Microsoft Building Footprint, Land Information New Zealand, and Copernicus Global Land Service data to improve distribution accuracy.



Population Density (Population Count Per Hectare)

Centroid Coordinates: 79982E, 454320N
Hex Grid: ~500m (Side length)
Data Source: Kontur Population Dataset (kontur.io/portfolio/population-dataset)

Projection: EPSG:28992
Name: Amersfoort / RD New -- Netherlands - Holland - Dutch
Unit: metre
Geodetic CRS: Amersfoort
Datum: Amersfoort
Ellipsoid: Bessel 1841

2.2 Data Processing

Hourly measurements from both official and community sensors were aggregated to daily frequency. Community sensor data was merged with the nearest official sensor data for corresponding dates. This synchronization of daily measurements enabled direct comparison. However, due to the non-standard temporal coverage of community sensors, a significant portion of official sensor data had to be dropped where matching data was unavailable.

2.2.1 Defining "Reliability"

Reliability was evaluated through a comprehensive analysis of the agreement between official and community sensor measurements. Scatter plots were generated to visualize the relationship, and the Pearson correlation coefficient (r) was computed to quantify the strength of the correlation. A higher r value indicated stronger agreement and reliability. Time-series graphs were also plotted to examine temporal trends, ensuring alignment of high and low pollution events across datasets. To enhance the evaluation, a composite reliability index was developed by integrating Pearson correlation, the number of observations, root mean square error, and the distance between sensors. This index provides a robust measure of reliability by accounting for both spatial proximity and the quality of long-term measurement consistency.

Reliability was defined as follows:

1. Pearson Correlation (r): Normalise Pearson Correlation between [0,1] using the below formula

$$r_{norm} = \frac{r+1}{2}$$

2. Number of Observations (n): The number of observations were normalised using min-max normalisation. More observations provide higher confidence in the reliability.

$$n_{norm} = \frac{n - n_{min}}{n_{max} - n_{min}}$$

3. Distance between sensors (d): The distance between sensors were normalised inversely to prioritise closer proximity.

$$d_{norm} = 1 - \frac{d - d_{min}}{d_{max} - d_{min}}$$

4. Root Mean Square Error (rmse): Root mean square error was computed between official & community measurements. Higher RMSE signifies lower reliability.

$$rmse_{norm} = 1 - \frac{MSE - MSE_{min}}{MSE_{max} - MSE_{min}}$$

5. Weighting Factors: A weighted sum of normalise factors was computed, with weights assigned to reflect their relative importance (for eg., for correlation, for observation, for distance, for RMSE)

$$Reliability\ Index = w_1 * r_{norm} + w_2 * n_{norm} + w_3 * d_{norm} + w_4 * rmse_{norm}$$

2.2.2 Placement Strategy

To identify optimal locations for new air quality sensors, a weighted scoring approach similar to the reliability index was employed, incorporating population, number of schools, and the existing official sensor network. Population data, represented in H3 hexagons with a 400m resolution, served as the baseline for population counts. School data was overlaid, and a point-in-polygon query was conducted to aggregate the number of schools within each grid cell. Additionally, the distance to the nearest official sensor was calculated from the centroid of each grid cell. The resulting dataset contained values for total population, school count, and distance to the nearest official sensor for each grid cell. Each factor was normalized using min-max normalization to scale values between 0 and 1. A composite placement score was then calculated by taking the weighted sum of these factors. The weights were assigned based on their relative importance: 0.5 for school count, 0.3 for population count, and 0.2 for distance to the closest sensor. The hexgrid was visualized based on the composite scores, with the four existing official sensors overlaid to evaluate current school coverage. Grid cells with high composite scores (> 0.5) were prioritized as candidate locations for six new official sensors. Cells with lower composite scores (< 0.5 & >0.3) were identified as suitable for community sensors to ensure broader spatial coverage.

2.1 Methodology Workflow

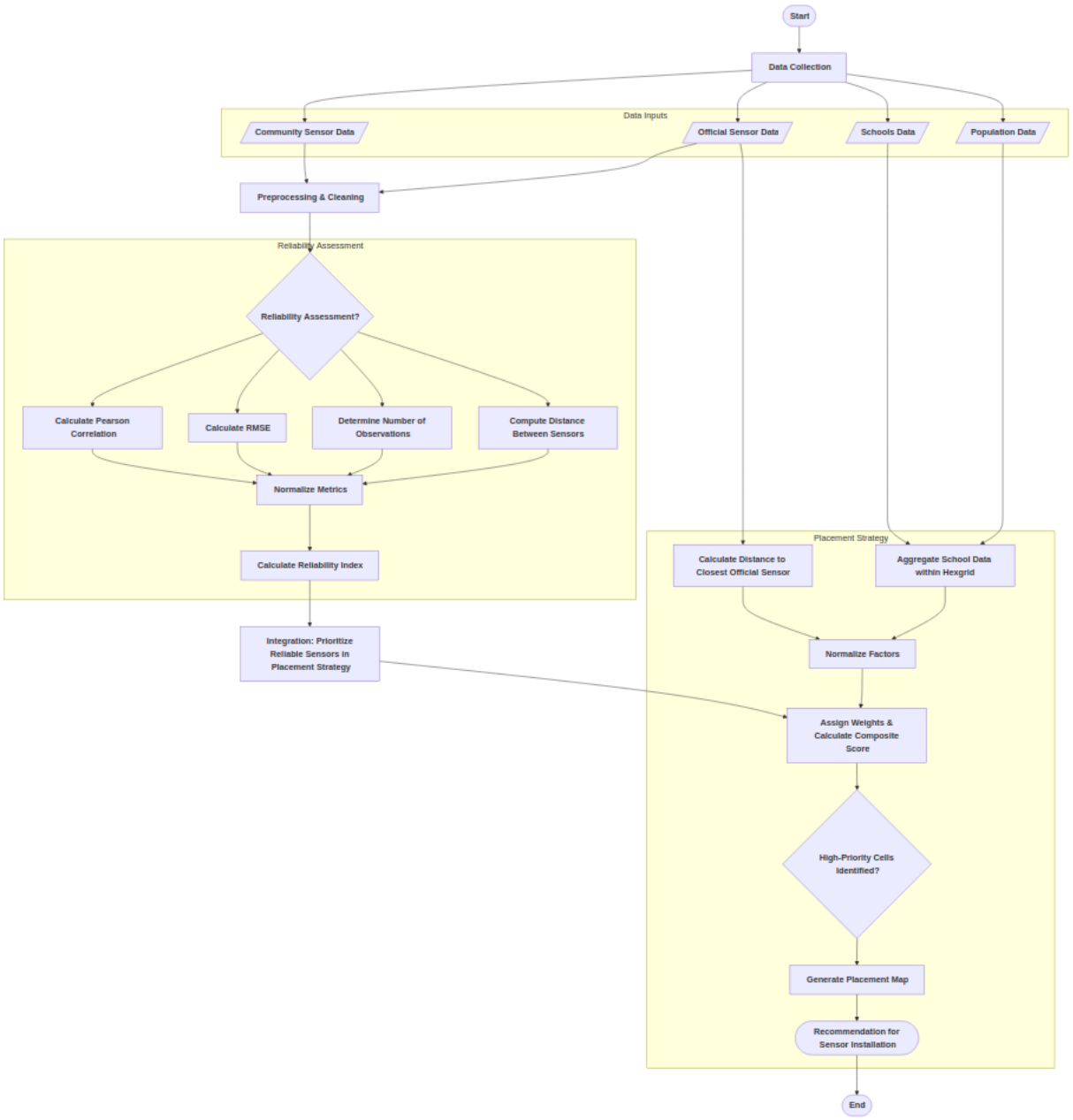


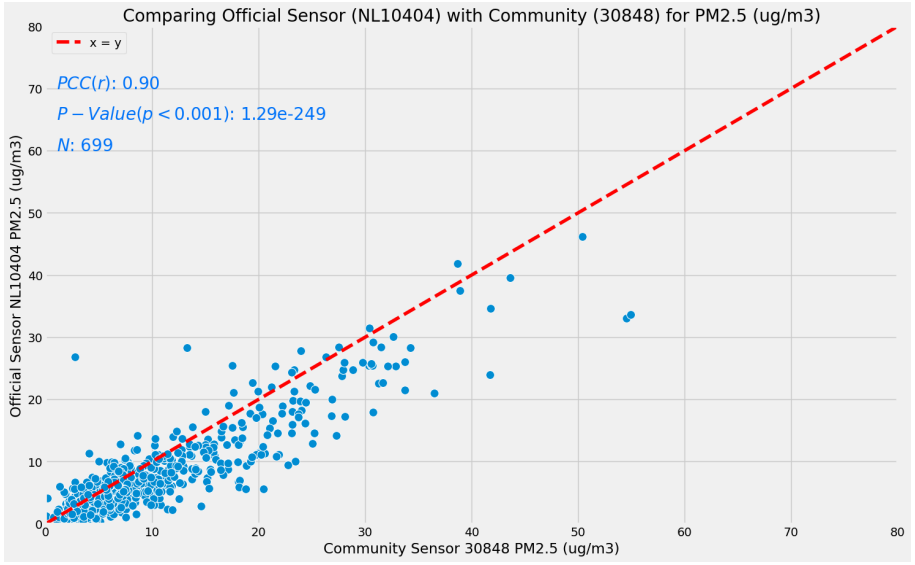
Figure 1 Methodology Flowchart

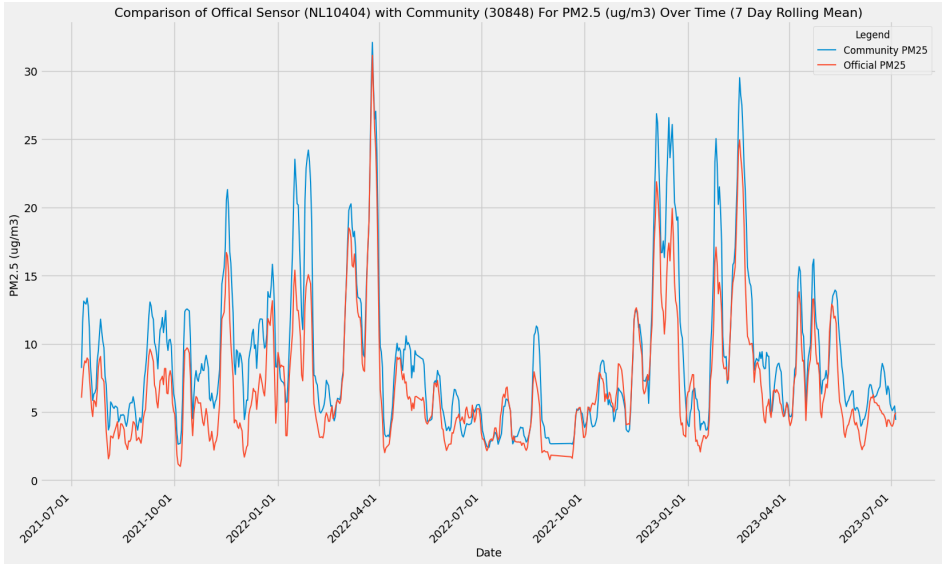
3. Results

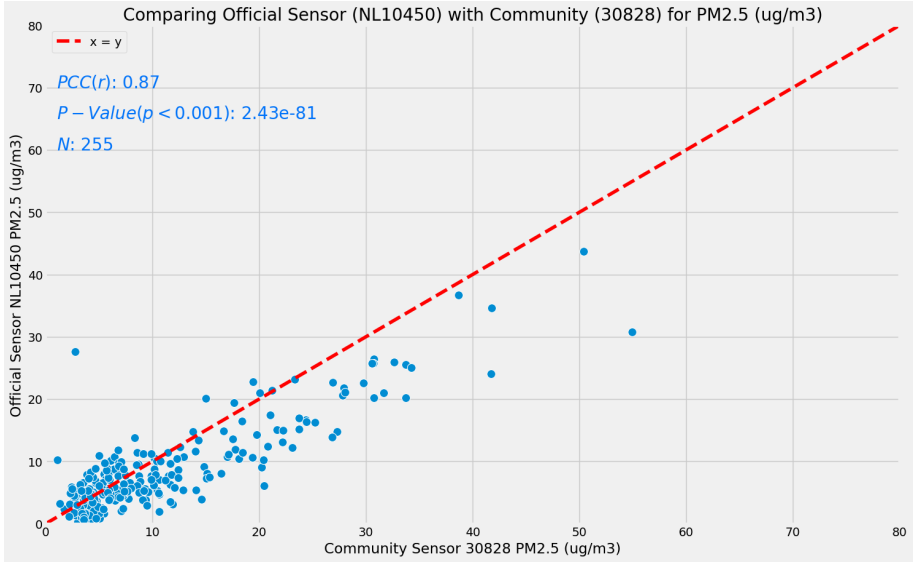
To evaluate the performance and reliability of community air quality sensors relative to official sensors, key metrics were analyzed, including mean, standard deviation, Pearson correlation coefficient, reliability index, and the number of observations. **Table 1** provides a summary of the comparison of $PM_{2.5}$ and PM_{10} concentrations measured by community and official sensors, highlighting variations in performance and data quality.

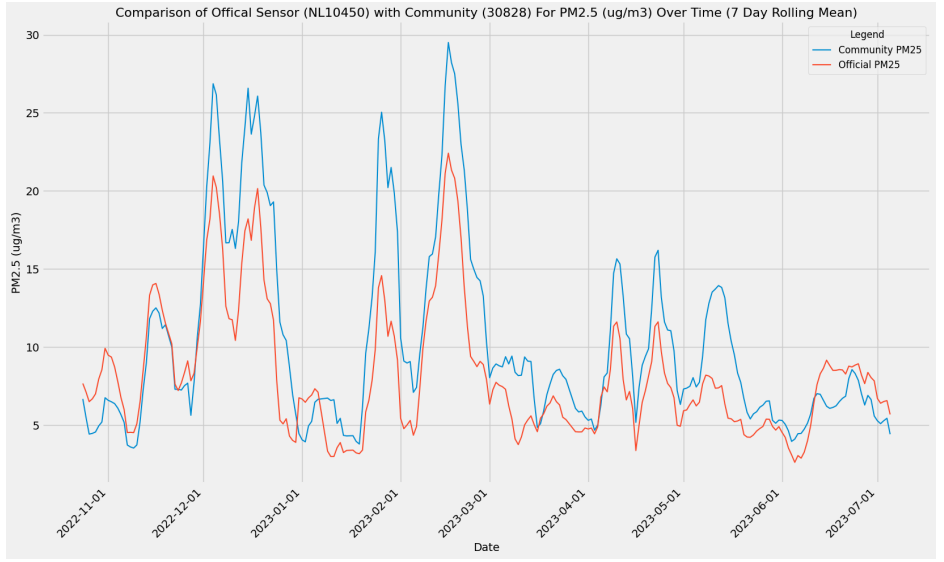
Pollutant	Official Sensor	Community Sensor	Number of Observations	Distance From Official Sensor(Km)	r(PCC)	Reliability Index	p<0.001	RMSE(ug/m3)
$PM_{2.5}$	NL10404	30848	699	0.586	0.90	0.91	1.29e-249	4.08
$PM_{2.5}$	NL10450	30828	255	1.240	0.87	0.64	2.43e-81	5.03
PM_{10}	NL10404	13869	457	1.188	0.84	0.73	5.70e-123	5.45
PM_{10}	NL10445	13871	364	0.704	0.91	0.76	6.15e-139	5.07
PM_{10}	NL10446	20437	104	1.641	0.86	0.53	3.62e-32	6.25
PM_{10}	NL10450	29987	41	0.317	0.91	0.66	4.58e-16	4.85

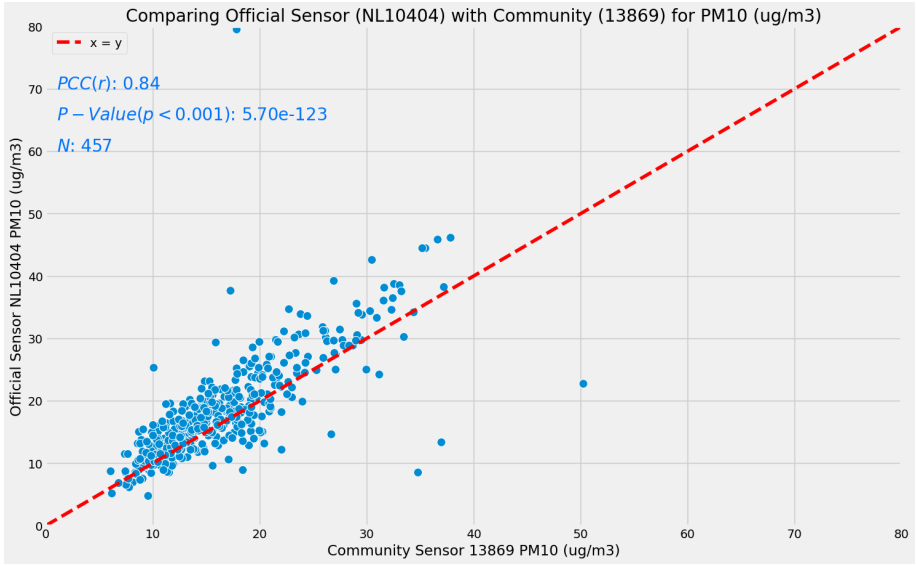
For $PM_{2.5}$, sensor pair **(NL10404, 30848)**—comprising an official and a community sensor—achieved a high reliability index of **0.91**, attributed to its strong correlation, closer proximity, higher number of observations, and lower RMSE. In contrast, pair **(NL10450, 30828)** had a lower reliability index of **0.64**, despite a similar correlation. The lower score was due to having half the number of observations and being twice as far from the official sensor.

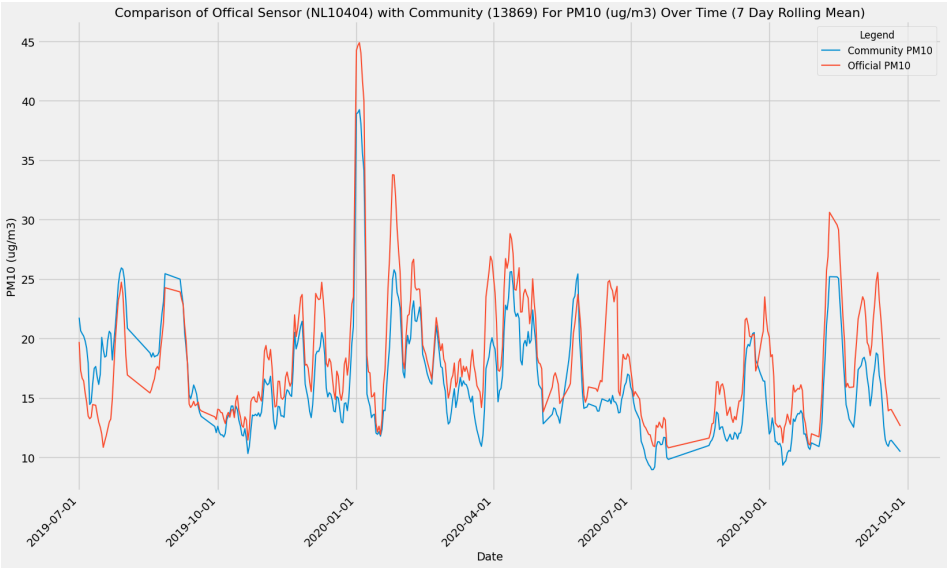

Comparing Official Sensor (NL10404) with Community (30848) for PM2.5 (ug/m3)
x = y
PCC(r): 0.90
P – Value(p < 0.001): 1.29e-249
N: 699
Official Sensor NL10404 PM2.5 (ug/m3)
Community Sensor 30848 PM2.5 (ug/m3)

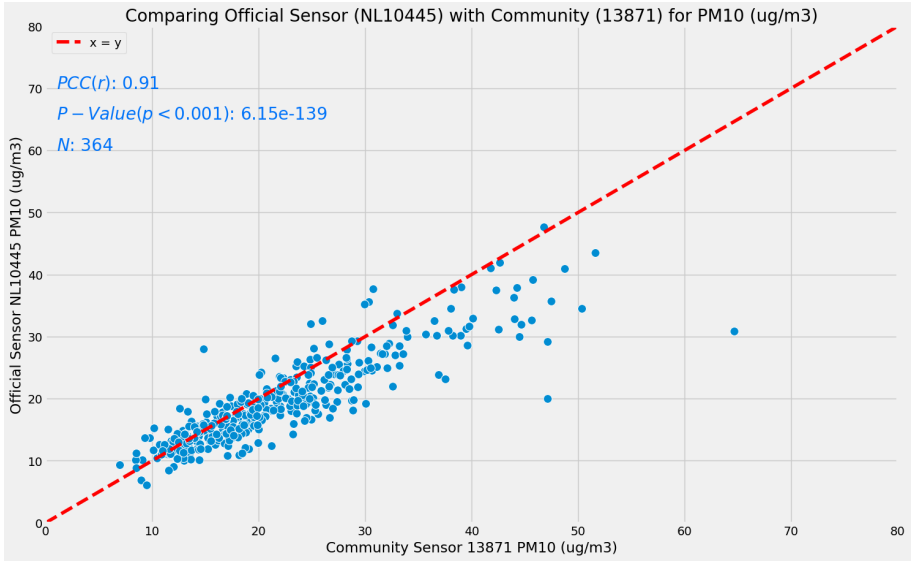

Comparison of Offical Sensor (NL10404) with Community (30848) For PM2.5 (ug/m3) Over Time (7 Day Rolling Mean)
Legend
Community PM25
Official PM25
PM2.5 (ug/m3)
Date


Comparing Official Sensor (NL10450) with Community (30828) for PM2.5 (ug/m3)
x = y
PCC(r): 0.87
P – Value(p < 0.001): 2.43e-81
N: 255
Official Sensor NL10450 PM2.5 (ug/m3)
Community Sensor 30828 PM2.5 (ug/m3)


Comparison of Offical Sensor (NL10450) with Community (30828) For PM2.5 (ug/m3) Over Time (7 Day Rolling Mean)
Legend
Community PM25
Official PM25
PM2.5 (ug/m3)
Date


Comparing Official Sensor (NL10404) with Community (13869) for PM10 (ug/m3)
x = y
PCC(r): 0.84
P – Value(p < 0.001): 5.70e-123
N: 457
Official Sensor NL10404 PM10 (ug/m3)
Community Sensor 13869 PM10 (ug/m3)


Comparison of Offical Sensor (NL10404) with Community (13869) For PM10 (ug/m3) Over Time (7 Day Rolling Mean)
Legend
Community PM10
Official PM10
PM10 (ug/m3)
Date


Comparing Official Sensor (NL10445) with Community (13871) for PM10 (ug/m3)
x = y
PCC(r): 0.91
P – Value(p < 0.001): 6.15e-139
N: 364
Official Sensor NL10445 PM10 (ug/m3)
Community Sensor 13871 PM10 (ug/m3)

For PM_{10}, a similar trend was observed. Sensor pair **(NL445, 13871)** achieved a reliability index of **0.76**, while **(NL10404, 13869)** scored slightly lower at **0.73**, attributed to being marginally farther, with lower correlation and higher RMSE. An interesting observation arose with pairs **(NL10446, 20437)** and **(NL10450, 29987)**. Despite being 1.64 km apart with 104 observations, the former scored **0.53**. Conversely, the latter, only 300 m apart but with just 41 observations, achieved **0.66**. This illustrates how distance, observation count, and RMSE influence the reliability index, helping identify sensors that may require calibration or further attention.

Time-series trends revealed another notable observation: for most sensor pairs, community sensors tended to overestimate concentrations compared to official ones. This discrepancy could stem from the higher sensitivity of low-cost sensors to their surroundings, necessitating additional calibration.

To address spatial gaps in air quality monitoring, a placement strategy was devised using a composite score based on **population density**, **school density**, and **proximity to existing sensors**. **Figure Y** visualizes optimal locations for six new official air quality sensors and 30 community sensors. High-priority grid cells with composite scores ≥ **0.5** were selected for official sensors, while cells with lower scores were deemed suitable for community sensors.

The placement map highlights high-priority areas such as the **Centrum region**, where high population and school densities coincide with a lack of existing sensor coverage. This strategy ensures enhanced monitoring in densely populated and vulnerable areas, particularly schools, which house children most at risk from air pollution.

4. Discussion

This section discusses the results of the case study and their interpretations.

4.1 Reliability Assessment

The reliability indicator was calculated using Pearson's correlation, RMSE, the number of observations, and the distance between sensors. The results showed significant variation across different sensor pairs, driven by disparities in the chosen input indicators. The analysis revealed patterns where sensors with strong correlation, low error metrics, proximity to official sensors, and minimal missing data were deemed highly reliable. For instance, in PM2.5 measurements, one of the chosen community sensors had a reliability index of 0.91, while the

other scored 0.64. Similarly, for PM10, the reliability index ranged from 0.76 to 0.53, even lower than for PM2.5. This demonstrates that the selected indicators effectively captured not only spatial (distance) and temporal contexts (long time series data) but also the accuracy of measurements (RMSE and correlation), lending trust and significance to the data quality. The proposed reliability index can be adapted to include additional environmental indicators that affect air quality, enhancing its robustness. Such reliable data can empower environmental bodies and governments to develop effective policies for improving air quality monitoring.

4.2 Placement Strategy

Optimal placement of air quality sensors is critical to improving spatial coverage and eliminating gaps in air quality monitoring. It also prevents redundancy and unintentional bias introduced by placing sensors too close to each other. This study presents a framework that utilizes a weighted sum of indicators—school density, population density, and proximity to existing sensors—to classify optimal locations for sensor placement. These indicators highlight the densest areas of society, where the majority of the population resides and schools, which house children most vulnerable to air pollution, are located. Based on the weighting mechanism, the algorithm identified areas with poor or no coverage from existing official air quality sensors. For example, areas near Centrum, which have the highest population and school density, are currently not covered by any official sensors. Similar patterns were observed across the municipality. The study proposes installing six additional official sensors in cells with the highest composite scores (≥ 0.5) and deploying 30 additional community sensors in surrounding cells with scores between 0.3 and 0.5. This framework provides a robust foundation for improving air quality monitoring while covering the majority of the population and schools. It can be further optimized and adapted by integrating other indicators, such as road networks and industrial versus residential areas.

4.3 Limitations

This study is not without its constraints. Temporal gaps and inconsistencies in the observations of community sensor data posed significant challenges during analysis. Many sensors either lacked data for extended periods or provided identical data without clarity about the source. The weighting scheme for the reliability index is based on predefined indicators with constant weights, which may introduce bias. Dynamic weight assignment based on spatial and temporal contexts could improve the framework. Similarly, normalization parameters might vary across regions and cities, which could limit generalizability. Finally, the spatial generalizability of the current strategy to cities with different population distributions and air quality characteristics remains untested.

5. Conclusion

This study demonstrates a comprehensive approach to improving air quality monitoring by integrating a reliability index and an optimized placement strategy. By utilizing key indicators, such as correlation, RMSE, the number of observations, and proximity, the reliability index effectively highlights the quality of sensor data. The placement strategy ensures efficient sensor deployment, addressing gaps in spatial coverage and prioritizing vulnerable populations.

Future work aims to scale and expand this proof of concept by:

1. Refining the reliability index with additional factors, such as weather data and terrain information.
2. Validating the placement strategy in new cities and regions.
3. Developing dynamic models for weight assignment.
4. Including additional factors, such as traffic patterns, industrial emissions, and time-of-day effects.
5. Extending the methodology to cover other key locations, such as hospitals and playgrounds.

By addressing these aspects, the framework can become a more adaptable and comprehensive solution for air quality monitoring, benefiting public health and urban planning efforts.

6. References

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Appendix

Data Management Plan

Table 1 Data Management Plan

Data Description	Format	Spatial Resolution	Data of Acquisition	CRS
Luchtmeet Air Quality Data	Vector Layer (.geojson)	-	31-12-2024	EPSG:28992
Samen Meten	Vector Layer (.geojson)	-	31-12-2024	EPSG:28992
Administrative Boundaries	Vector Layer (.geojson)	-	31-12-2025 (The Netherlands)	EPSG:28992
			31-12-2025 (The Hague)	EPSG:28992
Schools	Vector Layer (.geojson)	-	31-12-2025 (The Hague)	EPSG:28992
Kontur Population Data	Vector Layer (.gpkg)	400m	31-12-2025 (The Hague)	EPSG:28992